Lab 6 – Lab Reflection

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System/Project: Password Policy Checker – Code and Branch Coverage Analysis

1. What I Learned

In this lab, I learned the difference between code coverage and branch coverage and how test coverage can help identify parts of an application that have not been adequately exercised. I used pytest to execute tests and coverage.py to measure coverage.

2. Application Testing Experience

The application under test was a Password Policy Checker. Its functions validate password length and character types, identify common passwords, evaluate password strength, and suggest fixes for detected issues. I created separate test suites for code coverage and branch coverage.

3. Code Coverage Analysis

I expanded the code-coverage test suite to exercise the application's validation, strength evaluation, and suggestion functions. The recorded run passed all 17 tests and reported 100% code coverage for password_policy.py, with 47 statements and zero missed statements.

4. Branch Coverage Analysis

I ran the branch-coverage test suite with the --branch option. All 9 tests passed, and the report showed 100% statement coverage with 36 branches and zero partial branches. The coverage report did not list any missing branches, indicating that all reported branch paths were covered.

5. Challenges Faced

One challenge was distinguishing coverage of the test file from coverage of the application itself and ensuring the coverage command targeted password_policy.py. I also needed a separate branch-enabled run because ordinary code coverage does not measure branch outcomes in the same way.

6. Overall Reflection

Overall, this lab improved my understanding of automated testing and coverage measurement. Achieving 100% code coverage and 100% branch coverage in the recorded reports showed that the test cases exercised all reported statements and branch paths in the application. The screenshots provide evidence of these results.

7. Coverage Results Screenshots

Code Coverage Result

```
/content/L6_L7_39_password_policy
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-8.4.2, pluggy-1.7.0 -- /usr/bin/python3
cachedir: .pytest_cache
rootdir: /content/L6_L7_39_password_policy
plugins: anyio-4.15.1, typeguard-4.6.0, platformdirs-4.13.5, langsmith-0.14.12
collected 17 items

lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_valid_password PASSED [  5%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_password_too_short PASSED [ 11%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_password_too_long PASSED [ 17%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_common_password PASSED [ 23%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_missing_uppercase PASSED [ 29%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_missing_lowercase PASSED [ 35%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_missing_digit PASSED [ 41%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_missing_special_character PASSED [ 47%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_strong_password PASSED [ 52%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_good_password PASSED [ 58%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_weak_password PASSED [ 64%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_very_weak_password PASSED [ 70%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_suggest_fix_for_short_password PASSED [ 76%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_suggest_fix_for_common_password PASSED [ 82%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_suggest_fix_for_missing_character PASSED [ 88%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_suggest_fix_for_other_issue PASSED [ 94%]
lab6/CodeCoverageTest_PES1UG24AM221_application.py::test_no_fix_needed PASSED [100%]

============================== 17 passed in 0.10s ==============================
Name                       Stmts   Miss  Cover   Missing
--------------------------------------------------------
lab6/password_policy.py       47      0   100%
--------------------------------------------------------
TOTAL                         47      0   100%
```

Code Coverage Result — submitted execution output.

Branch Coverage Result

```
%cd /content/L6_L7_39_password_policy

!coverage erase
!coverage run --branch --include="*/password_policy.py" -m pytest -v lab6/BranchCoverageTest_PES1UG24AM221_application.py
!coverage report -m
```

```
/content/L6_L7_39_password_policy
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-8.4.2, pluggy-1.7.0 -- /usr/bin/python3
cachedir: .pytest_cache
rootdir: /content/L6_L7_39_password_policy
plugins: anyio-4.15.1, typeguard-4.6.0, platformdirs-4.13.5, langsmith-0.14.12
collected 9 items

lab6/BranchCoverageTest_PES1UG24AM221_application.py::test_length_branches PASSED [ 11%]
lab6/BranchCoverageTest_PES1UG24AM221_application.py::test_common_password_branches PASSED [ 22%]
lab6/BranchCoverageTest_PES1UG24AM221_application.py::test_character_type_branches PASSED [ 33%]
lab6/BranchCoverageTest_PES1UG24AM221_application.py::test_strength_branches PASSED [ 44%]
lab6/BranchCoverageTest_PES1UG24AM221_application.py::test_suggest_fix_empty_issues PASSED [ 55%]
lab6/BranchCoverageTest_PES1UG24AM221_application.py::test_suggest_fix_too_short PASSED [ 66%]
lab6/BranchCoverageTest_PES1UG24AM221_application.py::test_suggest_fix_common PASSED [ 77%]
lab6/BranchCoverageTest_PES1UG24AM221_application.py::test_suggest_fix_missing_character PASSED [ 88%]
lab6/BranchCoverageTest_PES1UG24AM221_application.py::test_suggest_fix_fallback PASSED [100%]

============================== 9 passed in 0.08s ===============================
Name                       Stmts   Miss Branch BrPart  Cover   Missing
----------------------------------------------------------------------
lab6/password_policy.py       47      0     36      0   100%
----------------------------------------------------------------------
TOTAL                         47      0     36      0   100%
```

Branch Coverage Result — submitted execution output.